✧ Triage Copilot

Classify, route, prioritise and flag risk for a new case.

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Model cards

M1 — Case type (A)

Approach summary-only text classifier
Inputs summary only
Split group-on-summary
Core accuracy **0.875** · macro_f1 **0.875** · weighted_f1 **0.875** · cohen_kappa **0.75**

M2 — Category (B)

Approach summary-only text classifier
Inputs summary only (tags excluded — leakage)
Split group-on-summary
Core accuracy **0.901** · macro_f1 **0.89** · weighted_f1 **0.902** · cohen_kappa **0.889**
Signature confidence_gap **0.24**

M3 — Priority — direct (C)

Approach direct priority predictor
Inputs summary + plan_tier + tenure + channel
Split time (+random ref)
Core accuracy **0.553** · macro_f1 **0.587** · weighted_f1 **0.55** · cohen_kappa **0.379**
Signature within_1_level **0.93** · quadratic_kappa **0.454** · macro_f1_random_split **0.671**

M3 — Priority — review check (D)

Approach priority double-check (ordinal, flag mismatches)
Inputs summary + case_type + channel (no sla_target, no plan proxy)
Split time
Core accuracy **0.444** · macro_f1 **0.502** · weighted_f1 **0.44** · cohen_kappa **0.241**
Signature within_1_level **0.926** · quadratic_kappa **0.313** · flag_rate **0.074**

M5 — Suggested route (F)

Approach summary-only router with review gate
Inputs summary only
Split time
Core accuracy **0.848** · macro_f1 **0.805** · weighted_f1 **0.854** · cohen_kappa **0.776**
Signature pct_needs_review **0.163** · acc_when_confident **0.93**

operations merged into support; review gate at 0.5

M4 — Escalation risk (E)

Approach escalation risk, recall-targeted threshold
Inputs summary + category + channel + month + tenure (no tags — leakage)
Split time 70/15/15
Core accuracy **0.934** · macro_f1 **0.793** · weighted_f1 **0.94** · cohen_kappa **0.588** · ROC_AUC **0.96** · PR_AUC **0.74**
Signature baseline_PR_AUC **0.07** · threshold **0.35** · recall_pos **0.778** · precision_pos **0.519** · min_recall_target **0.8**

Normalised held-out metrics

Model	Predicts	n_test	accuracy	macro_F1	weighted_F1	cohen_kappa	ROC_AUC	PR_AUC	baseline_PR_AUC	baseline_acc
M1	Case type	344	0.875	0.875	0.875	0.75				0.507
M2	Category	344	0.901	0.89	0.902	0.889				0.147
M3	Priority — direct	257	0.553	0.587	0.55	0.379				0.426
M3	Priority — review check	257	0.444	0.502	0.44	0.241				0.426
M5	Suggested route	257	0.848	0.805	0.854	0.776				0.395
M4	Escalation risk	257	0.934	0.793	0.94	0.588	0.96	0.74	0.07	0.866

Inspect a held-out case — predicted vs actual

Model: M1 — Case type

Held-out case: ND-2025-000400 — Need a VAT receipt for the last Orbit ID subscription p

"Need a VAT receipt for the last Orbit ID subscription payment cheers"

Model	M1 — Case type
Predicted	service_request (71%)
Actual	service_request
Result	match

Held-out case (group-on-summary split) — never seen in training.

Confusion matrix & per-class report

Model: M1 — Case type

rows = actual · columns = predicted

	incident	service_re
incident	154	22
service_re	21	147

class	prec	rec	F1	n
incident	0.88	0.88	0.88	176
service_request	0.87	0.88	0.87	168

Browse held-out cases

Model: M1 — Case type

Sort by result: Case order

344 held-out cases · 88% correct · group-on-summary split (never seen in training)

CASE	SUMMARY	PREDICTED	CONF	ACTUAL	RESULT
ND-2025-000400	**Need a VAT receipt for the last Orbit ID subscription payment cheers**	service_request	71%	service_request	correct
ND-2025-000402	**The invoice amount for the Enterprise tier doesn't look right for March**	service_request	72%	incident	wrong
ND-2025-000404	**Where do I download the VAT invoice for my records?**	service_request	90%	service_request	correct
ND-2025-000409	**I've been charged twice for the same connector add-on**	incident	74%	incident	correct
ND-2025-000415	**VAT invoice missing for Feb please sort**	service_request	91%	service_request	correct
ND-2025-000420	**Copy of VAT receipt thanks**	service_request	92%	service_request	correct

Training notebooks (the real notebooks used to build each model)

⬇ m1_case_type · ⬇ m2_category · ⬇ m3_priority_direct · ⬇ m3_priority_review · ⬇ m5_route · ⬇ m4_escalation

⬇ similar_cases · ⬇ tag_suggester